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Annals of Operations Research

ISSN 0254-5330

Ann Oper Res
DOI 10.1007/s10479-020-03521-y

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Annals of Operations Research
https://doi.org/10.1007/s10479-020-03521-y

S.I.: DATA MINING AND DECISION ANALYTICS

A further analysis of robust regression modeling and data mining corrections testing in global stocks

John B. Guerard Jr.[1] · **Ganlin Xu**[1,2] · **Harry Markowitz**[1,3]

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Abstract
In this analysis of the risk and return of stocks in global markets, we build a reasonably large number of stock selection models and create optimized portfolios to outperform a global benchmark. We apply robust regression techniques, LAR regression, and LASSO regression modeling to estimate stock selection models. Markowitz-based optimization techniques is used in portfolio construction within a global stock universe. We apply the Markowitz–Xu data mining corrections test to a global stock universe. We find that (1) robust regression applications are appropriate for modeling stock returns in global markets; (2) weighted latent root regression robust regression techniques work as well as LAR and LASSO-Regressions in building effective stock selection models; (3) mean–variance techniques continue to produce portfolios capable of generating excess returns above transactions costs; and (4) our models pass several data mining tests such that regression models produce statistically significant asset selection for global stocks. Recent Sturdy-Regression modeling technique may offer the greatest potential for further research for statistically based stock selection modeling.

Keywords Portfolio optimization · Data mining · Testing · I/B/E/S forecasts

Earlier versions were presented at the INFORMS meeting, November 2015, in Philadelphia, PA, the Wharton–Jacobs and Levy Spring Forum, May 2016, the Conference on Statistical Learning and Data Science, the University of North Carolina, June 2016, Chapel Hill, and International Symposium of Forecasting, Santander, Spain, June 2016. The comments on earlier versions by Edward Leamer, Andrew Lo, Jose Menchero, Anureet Saxena, Jinghua Yan, and Laurence Wormald are gratefully acknowledged. Any remaining errors are the responsibility of the primary author.

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Published online: 07 February 2020

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1 An introduction, organization, and desired set of results and recommendations of our analysis

Markowitz (1952) created modern portfolio selection by developing an efficient frontier on which optimal portfolios achieved the maximum return for a given level of risk or the minimum risk for a given level of return. The set of optimal portfolios, called efficient portfolios, can be formulated as solutions to a parametric mean–variance quadratic optimization problem with return-risk tradeoff as parameter. Markowitz (1959) invented a fast and practical algorithm (critical line algorithm) to find all the efficient portfolios. The inputs to find efficient portfolios are expected returns, covariance–covariance among the securities, and constraints.

Modern portfolio theory is concerned with identifying statistically significant expected return models, constraints, and variance–covariance models. We will focus on the expected return model in this paper and use commercially available variance–covariance model. The general form of expected return model is

$$\mu_{t+1} = H\left(y_{t,1}, \ldots, y_{t,P}\right)$$

where μ_{t+1} is the N-dimensional expected return vector for t + 1 period, and $y_{t,p}(1 \le p \le P)$ are P observable or calculable N-dimensional firm specific features vector at the end of period t. N is the number of investable securities for period t + 1. The loss function for period t + 1 is

$$L\left(R_{t+1}, \mu_{t+1}\right)$$

where R_{t+1} is the observed return vector of N-dimensions. We will discuss in depth how to choose the P firm feature variables, the truth function form of H, and the loss function form of L. The firm specific feature variables must satisfy two criteria: (1) the predictive information content must be statistically significant and persistent; and (2) statistical techniques can be used to extract the information to produce the higher return stock portfolios. Bloch et al. (1993) created and tested a large set of fundamental variables, including earnings, book value, cash flows, and sales, in Japan and the U.S. over a period of approximately 15 years. In their published research, Bloch et al. (1993) found a linear model with eight fundamental variables, weighted L^2 loss function, and robust latent root regression worked best. Portfolio management seeks to maximize the Geometric Mean, Sharpe Ratio, the portfolio excess return divided by the portfolio standard deviation, and the Information Ratio, the portfolio excess return divided by its tracking error.

Bloch et al. (1993) ranked the portfolios based on the Sharpe Ratio and designated WLRR as the best model after tested more than hundreds of models. Is WLRR really the best model or just lucky one? Markowitz and Xu (1994) developed a data mining corrections test (brief description in section six) which established that WLRR's return were significantly higher than all model's average return. The data mining test establishes statistically that the excess returns were not due to chance and estimate what fraction of the excess returns could be expected in the future. Guerard et al. (2015) confirmed the post-publication stock selection and portfolio construction models used in Bloch et al. (1993) and introduced three levels of testing for statistical significance. Moreover Guerard et al. (2015) verified the introduction of earnings forecasting models in Guerard et al. (1997) and the price momentum work of Fama and French (1995).

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In this analysis, we update and extend the Bloch et al. (1993) study to create a set of global portfolios over the 2003–2015 period with following questions in mind:

1. Does the original expected model still work after its publication? Using post publication period and different global universe is the ultimate validation of our expected return modeling methodology.
2. We have extended the predictive feature variables (univariate model) to thirty-one. Instead of choosing eight or ten variables for all simulation periods, Can the recent best subset variable selection methods, such as Least Angle Regression and LASSO Regression produce a better portfolio on overall period than the outlier-adjusted and multicollinearity-adjusted model in Bloch et al. (1993)?

In this introductory section, we provide a guide for the reader to follow our updated and enhanced testing of stock selection modeling, portfolio construction and management, and statistical testing of portfolio returns, including data mining tests. We report the extended predictive variables include forecasted earnings acceleration and price momentum variables modeled in Guerard et al. (2012, 2013, 2014, 2015, 2018a, b). The expected returns are input to an optimization system which uses a variation of traditional mean–variance and mean–variance tracking error at risk portfolio construction techniques. In Sect. 2, we introduce the reader to an established stock selection model that has been reported, and confirmed, to identify statistically significant stock selection models during the 1974–1990 and 1997–2011 time periods. In Sect. 3, we briefly introduce the reader to financial variables in stock selection modeling and regression modeling. We describe the database for our analysis. The reader is introduced to Level I statistical testing of stock selection variables. The regression models used to create expected returns combine well-established fundamental factors, such as earnings, book value, cash flow, and sales, forecasted earnings acceleration, and price momentum factors. In Sect. 3, we discuss the data used in our analysis which covers the 2002–May 2015 time period. In Sect. 4, the reader is introduced to Markowitz portfolio construction of mean–variance efficient portfolios. The statistically significant factors of Sect. 3 are used as portfolio tilt variables and in the multiple-factor portfolio construction models of Sect. 4. In Sect. 5, we report portfolio simulation results and we introduce the reader to the Markowitz and Xu (1994) data mining corrections tests, and more modern data mining tests. In Sect. 6, we report our summary and conclusions.

We report that (1) robust regression applications continue to be appropriate for modeling stock expected returns in global markets; (2) mean–variance techniques continue to produce portfolios capable of generating excess returns above transactions costs; and (3) our best models pass data mining tests such that advanced regression models produce statistically significant asset selection; and (4) LAR and LASSO Regressions does not improve our robust regression model for stock selection in terms of geometric mean.

2 A review fundamental variables and regression-based expected returns modeling

Stock selection and security analysis is not new venue of research. Graham and Dodd (1934) and Williams (1938) developed the framework for stock analysis well over 75 years ago, formulating the low price-to-earnings multiple, P/E, strategy, in which no investor should purchase a stock with a P/E that exceeded 50% more than the market P/E. In

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1991, the Markowitz's Global Portfolio Research Department (GPRD) tested a set of 16 financial variables, including the low P/E model, and composite stock selection models over a 15-year period in Japan and the U.S. The Bloch et al. (1993) universes were large Japanese and U.S. stocks. The Japanese stocks were the non-financial First Section of the Tokyo Stock Exchange. The U.S. stocks were the largest 1000 market-capitalized stocks at the point in time. The Markowitz team used relative variables, defined as the ratio of the absolute fundamental variable ratios divided by the 60-month averages of the fundamental variable ratios. Guerard et al. (2012, 2015) updated and expanded the 8-factor U.S. stock selection model in Bloch et al. (1993) to include forecasted earnings acceleration and price momentum variables. There is a very large and growing literature of financial modeling, see Ramnath et al. (2008) and Harvey et al. (2016). For our analysis, we introduce the reader the Guerard et al. (2015) model which we use as the primary model of stock selection.

The 10-factor stock selection model estimated is:

$$TR_{t+1} = a_0 + a_1 EP_t + a_2 BP_t + a_3 CP_t + a_4 SP_t + a_5 REP_t + a_6 \mathrm{RBP_t} + a_7 RCP_t + a_8 RSP_t + a_9 CTEF_t + a_{10} PM_t + e_t \quad (1)$$

where TR, Total Security Returns in month t + 1;

EP, $\left[\text{earnings per share}\right] / \left[\text{price per share}\right]$

$= \text{earnings} - \text{price ratio};$

BP, $\left[\text{book value per share}\right] / \left[\text{price per share}\right]$

$= \text{book} - \text{price atio};$ CP, $\left[\text{cash flow per share}\right] / \left[\text{price per share}\right]$

$= \text{cash flow} - \text{price ratio};$

SP, $\left[\text{net sales per share}\right] / \left[\text{price per share}\right]$

$= \text{sales} - \text{price ratio};$

REP, $[\text{current EP ratio}] / \left[\text{average EP ratio over the past 5 years}\right];$

RBP, $[\text{current BP ratio}] / \left[\text{average BP ratio over the past 5 years}\right];$

RCP, $[\text{current CP ratio}] / \left[\text{average CP ratio over the past 5 years}\right];$

RSP, $[\text{current SP ratio}] / \left[\text{average SP ratio over the past 5 years}\right];$

$CTEF$, Forecasted Earnings Acceleration

$=$ average of consensus earnings per share forecast, revisions and breadth;

PM, Price Momentum $= PM71 = Price_{t-1} / Price_{t-7;}$

and e, randomly distributed error term.

Bloch et al. (1993) reported, and Guerard et al. (2015) confirmed the following results, post-publication that: (1) the Beaton–Tukey bisquare regression and multicollinearity-adjusted, denoted as WLRR, regression-based expected returns outperform equally-weighted models or ordinary least squares-weighted expected returns; (2) the risk model based on historical daily returns is an effective input to mean–variance optimization analysis; (3) optimization with turnover constraint enhanced portfolio returns; and (4)

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survivor-biased free stock universes were important, but not necessary for model development; and (5) modeling results must be tested using a data mining corrections test.[1] The model in Eq. (1) was referred to as a "Public", not a proprietary model in Bloch et al. (1993). This analysis continues that concept of the research that we share publicly.[2] We incorporate prior belief (all these feature variables should have positive coefficient) by normalizing the estimated a_t. The normalization process of a_t takes two steps. First, we set non-positive and non-statistically significant components to be zero. The remaining positive significant components are rescaled so that they add up to 1. Second, we average the normalized coefficients from past 12 months to get coefficients for forecasting. The 12-month smoothing is consistent with the four-quarter smoothing in Bloch et al. (1993). While EP and BP variables are significant in explaining returns, the majority of the forecast performance is attributable to other model variables, namely the relative earnings-to-price, relative cash-to-price, relative sales-to-price, price momentum, and earnings forecast variables.[3] The GLER model is very similar to the 12-factor model of Haugen and Baker.[4]

Practitioners on Wall Street test a very large number of factors or models. We test thirty-one variables or composite models in this analysis, many of which are not statistically significant all the time but are tested to reflect what is discussed in industrial publications; see Subramanian et al. (2018). Subramanian and her team test

[1] Robust regression is useful in analyzing data plagued by the presence of a "large number" of residuals from ordinary least squares regressions (OLS) that are outside of the 95% confidence interval. The reader is referred to Bloch et al. (1993) for a detailed explanation and rationale for robust regression, which was referred to as WLRR in the article. WLRR outperformed OLS in Sharpe Ratios and Geometric Means in the Bloch et al. analysis. Statisticians refer to the underlying robust regression model used in this study as the Beaton–Tukey bisquare weighting scheme, see Maronna et al. (2006, 2019) and Dhrymes (2017). Multicollinearity was a significant problem, given the presence of book value and sales, earnings and cash flow, and the variable interdependencies. Farrar and Glauber (1967), and Leamer (1973, 1978) addressed multicollinearity issues prior to Gunst and Mason (1980). The WLRR techniques eliminates the eigen roots with eigen values ≤ 0.30 and eigen vectors with first element ≤ 0.10 and is similar to the variance inflation regression technique of Lin et al. (2011).

[2] Guerard et al. (2013) estimated a Global Model, GLER, using Eq. (1) and the FactSet database for global securities during the January 1999–December 2011 period. In the world of business, one does not access academic databases annually, or even quarterly. Most industry analysis uses FactSet database and the Thomson Financial (I/B/E/S) earnings forecasting database. Guerard et al. (2013) estimated tracking error at risk portfolio (MVTaR) model for the 7500 largest securities, in terms of market capitalization, for stocks Thomson Financial and FactSet databases, some 46,550 firms in December 2011, and 64,455 stocks in December 2013. In the Guerard et al. (2015) earnings forecasting study used both APT and Axioma World-wide Statistical Risk Models. We use data only as it is known (or more exactly, our portfolios are tested out-of-sample). USER and GLER are the "Public Forms" of the McKinley Capital Management models for U.S. and Global stocks, respectively. Excess returns were higher in non-US than U.S. markets, see Deng and Min (2013) and Guerard et al. (2018a, b).

[3] The CTEF and PM variable weights are large relative to the first eight factor eights. The relative variables are "growth" variables such that both the Markowitz model and GLER models plot in the growth boxes of the Zephyr style report. The first four factors of GLER are value factors. The weighting results are extremely consistent with McKinley Capital Management being a Global Growth specialist. The CTEF and PM variables accounted for 40 percent of the weights in the GLER Model.

[4] Haugen (2001) continued the treatment of the Graham and Dodd variables in his Modern Investment Theory. Haugen and Baker (1996, 2010) examined 12 of the most important factors in the U.S. equity markets and in Germany, France, Great Britain, and Japan. The book-to-price, earnings-to-price, sales-to-price, and cash flow-to-price variables were among the highest mean payoff variables in the respective countries. Haugen and Baker (28) published a paper in the Guerard volume to honor Harry Markowitz which updated their models and completely demolished the case for efficient markets, in the eyes of the primary author. Our analysis and the work of Haugen do not report evidence that is consistent with the work of Fama and French (1992, 1995, 2008), and the dominance of the firm size and book-to-price variables in stock selection modeling.

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approximately 70 variables in the U.S. every quarter. The authors regularly review the Subramanian et al. (2018) work and variables. Many of our additional variables, beyond the set of 16 variables in Bloch et al. (1993) tested in this study, are suggested and tested in Subramanian et al. (2018), Levy (1999, 2012), Elton et al. (2007), and Harvey et al. (2016).

This paper differs from Guerard et al. (2015) in three aspects. First, the factors used in Eq. (1) have been increased from ten to thirty-one factors. Second, two new variable selection methods (Least Angle Regression and LASSO regression) are investigated. Third, several different data mining correction tests are calculated.

3 An initial model building criteria for stock selection models: data and initial variables tests of predictive power

Guerard et al. (2015) documented three levels of testing for effective stock selection and portfolio construction. One should first identify firm specific variables that are statistically associated with stock returns. The thirty-one variables tested here, together with composite variables, are listed as factor in Table 1. One generally estimates information coefficients (ICs) and excess (positive active) returns from univariate ranked decile portfolios for variables to be considered in possibly including in a composite model of stock returns.[5] We refer to the IC test as a Level I test. We use the Guerard et al. (2013, 2015) database of global stocks for the 1997–2011 time period, which we extend to 2015 to model the post-Global Financial Crisis time period. The global database was created such that each stock was covered by at least two I/B/E/S/analysts. Guerard et al. (2015) built expected returns using the 10-factor model presented in Eq. (1). The number of stocks grows to approximately 16,000 during the period 2003–2011. The universe was restricted to global stocks that were covered by at least two analysts, which reduced the number to approximately 7000–8000 stocks. One can survey the academic and practitioner literature, as we have done, and calculate information coefficients for the equity universe within which one seeks to manage assets. Information coefficients, ICs, are estimated by ranking strategies and the subsequent monthly returns. We estimate 1-month IC in this analysis. The benchmark is the Morgan Stanley Capital International (MSCI) Index-constituent data for the All Country World Growth Index (ACWG) for the 2003–5/2015 time period.[6] We test thirty-one fundamental, expectations, price momentum variables, and composite models of expected returns described in Guerard et al. (2015). We selected these variables because they are the most relevant and predictive variables in global growth stock universes. There is strong support for the earnings expectations variables and the fundamental variables (particularly earnings and cash flow). An objective examination of the reported ICs and annualized spread return numbers, as shown in Table 1 for the entire time period, leads one to identify CTEF as leading univariate variable. A regression weighting, WLRR of the GLER factors, using the weighted principal components regression discussed by Guerard et al. (2013), produces statistically significant high ICs. The predictive power of our WLRR model is virtually

[5] See Jacobs and Levy (1988) and Ziemba and Schwartz (1993) for different approaches to variable selection.

[6] Guerard et al. (2018a, b) tested 30-plus variables in six different U.S. and global stock universes.

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Table 1 Investment factors spread and information coefficients overview

12/31/2002 to 5/29/2015									
Universe return: 1.24									
Benchmark return: 0.74									
Factor	Annualized spread return	IC	IC T-Stat	IC SD	Annualized return quintiles				
					1	2	3	4	5
LAR	23.98	0.059	5.08	0.09	32.03	18.00	13.74	10.35	6.33
WLRR	22.31	0.056	4.86	0.09	31.03	16.93	14.26	11.38	6.70
CTEF	20.30	0.055	4.71	0.06	28.26	19.63	15.10	10.58	6.69
CTEFROIC	18.40	0.054	4.64	0.07	27.86	19.35	15.47	9.78	7.70
FY2RV3	12.95	0.041	3.56	0.07	25.40	16.97	15.63	11.02	10.34
PMTREND	14.35	0.039	3.41	0.11	23.98	19.03	15.59	12.77	7.95
BR2	10.63	0.037	3.15	0.05	22.82	16.32	17.73	10.31	11.57
FY1RV3	10.01	0.035	3.03	0.07	23.36	18.67	14.33	11.49	11.46
BR1	10.56	0.035	2.96	0.05	23.00	16.67	17.59	10.51	11.29
FEP1	12.30	0.032	2.73	0.09	26.19	14.79	13.38	12.37	12.49
ALPHA	9.38	0.031	2.69	0.12	21.38	16.97	17.58	13.59	9.26
FEP2	12.21	0.030	2.60	0.10	25.29	16.05	13.60	11.90	12.26
CP	10.50	0.028	2.43	0.08	23.75	17.10	14.62	12.44	11.30
DP	10.55	0.027	2.37	0.09	24.99	15.10	13.57	12.43	12.12
EP	9.70	0.023	2.01	0.08	25.11	13.91	13.00	13.61	13.53
STDEV	− 0.81	0.022	1.92	0.15	19.35	13.99	14.47	15.50	14.98
PM71	5.05	0.020	1.71	0.13	19.94	15.88	16.52	14.66	11.60
ROE1YR	1.24	0.017	1.42	0.06	15.90	18.34	16.82	14.48	13.58
ROIC	2.84	0.016	1.39	0.07	17.47	18.96	15.14	14.52	13.08
SP	5.48	0.015	1.32	0.09	17.79	19.99	16.57	13.25	11.53
ROA1YR	− 1.07	0.015	1.25	0.07	15.61	15.29	14.71	17.85	15.63
ROA3YR	− 0.25	0.014	1.19	0.06	14.86	15.57	15.42	19.26	14.02
ROA5YR	− 0.84	0.014	1.16	0.06	15.28	15.39	15.48	17.75	15.21
ROE3YR	0.77	0.013	1.16	0.06	14.91	18.42	17.39	15.09	13.31
NCSR	− 0.29	0.013	1.09	0.07	16.45	16.10	17.11	12.26	15.22
ROE5YR	0.85	0.012	1.07	0.06	14.78	18.54	18.01	14.64	13.20
DI	0.30	0.004	0.38	0.09	15.83	15.59	15.52	16.30	15.64
ES	2.82	0.004	0.36	0.07	19.55	15.42	14.32	14.56	15.15
CSR	1.25	0.004	0.34	0.09	17.65	13.85	− 0.84	15.88	19.35
BP	7.45	0.004	0.31	0.10	22.56	14.22	14.43	13.75	14.09
PM1	− 3.62	0.003	0.32	0.10	13.62	14.84	16.49	15.81	18.04
DR	− 3.91	0.002	0.14	0.05	15.16	15.38	13.98	12.88	20.17
NDR	− 1.29	− 0.004	− 0.34	0.08	15.41	15.96	16.36	15.32	15.96
CSI	− 0.09	− 0.011	− 0.92	0.10	14.66	14.55	12.08	16.09	14.79

EP, earnings per share/price per share; BP, book value per share/price per share; CP, cash flow per share/price per share; SP, sales per share/price per share; DP, dividends per share/price per share; PMTrend, price momentum with market effect removed; PM71, price momentum as $Price_{t-1}/Price_{t-7}$; FEP1, 1-year-ahead forecast earnings per share/price per share; FEP2, 2-year-ahead forecast earnings per share/price per share; FY1RV3, 1-year-ahead forecast earnings per share monthly revision/price per share; BR1, 1-year-

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Table 1 (continued)

ahead forecast earnings per share monthly breadth; BR2, 2-year-ahead forecast earnings per share monthly breadth; ROE_1Yr, 1-year return on equity; ROE_5Yr, 5-year return on equity; ROA_1Yr, 1-year return on total assets; ROA_3Yr, 3-year return on total assets; ROA_5Yr, 5-year return on total assets; CTEF, equally-weighted FEP1, FEP2, BR1, BR2, RV1, and RV2; LAR, Least Angle Regression of (EP, BP, CP, SP, REP, RBP, RCP, RSP, CTEF, and PM); WLRR, Weighted Latent Root Regression of (EP, BP, CP, SP, REP, RBP, RCP, RSP, CTEF, and PM); ROIC, return on invested capital; CSR, common stock repurchased; CSI, common stock issued; NCSR, net common stock repurchased; DR, debt repurchased; DI, debt issued; NDR, net debt repurchased; ES, corporate exports. = Dividends + Interest Paid + Net Equity Repurchased + Net Debt Repurchased; CTEFROIC, Proprietary-Weighted CTEF

identical to the application of Least Angle Regression (LAR) and LASSO-Regression in the Level I expected return modeling test.[7]

Level II and III tests of stock selection and portfolio construction and management require the estimation of efficient portfolios. Let us briefly review Markowitz efficient portfolio construction.

4 Constructing mean–variance efficient portfolios

The portfolio expected return, denoted by $E(R_{\mathrm{P}})$, is calculated by taking the sum of the security weight multiplied by their respective expected return:

$$E(R_{\mathrm{P}}) = \sum_{i=1}^{N} w_i E(R_i) \tag{2}$$

The portfolio risk (variance) is the sum of the weighted securities' covariance:

$$\sigma_{\mathrm{p}}^2 = \sum_{i=1}^{N} \sum_{j=1}^{N} \mathrm{w}_i w_j \sigma_{ij} \tag{3}$$

where N is the number of candidate securities, w_i is the weight for security i such that $\sum_{i=1}^{N} w_i = 1$, indicating that the portfolio is fully invested, and $E(R_i)$ is the expected return for security i.

The modeling of variance and covariance can be done by factor model too. The total excess return (relative to risk free rate) for a multiple-factor model (MFM) in the Rosenberg methodology for security j, at time t, may be written like this:

$$\mathrm{R}_{\mathrm{j,t}} = \sum_{\mathrm{k}=1}^{\mathrm{K}} \beta_{\mathrm{jk}} \tilde{\mathrm{f}}_{\mathrm{k,t}} + \tilde{\mathrm{e}}_{\mathrm{j,t}} \tag{4}$$

[7] Efron et al. (2004) introduce LAR to the reader by discussing automatic model-building algorithms, including forward selection, all subsets, and back elimination. LAR is a variation of forward selection; LAR creates a regression model, one covariate at a step, such that after K steps, only K of the $\hat{\beta}_j$s are non-zero. The K corresponding independent variables are the best K-member subset model. In our application, K is selected with highest Akaike Information Criterion (AIC). The K and selected sub-model may differ for each period. See Tibshirani (1996) for LASSO discussion. One of the better books on statistical modeling using LAR and LASSO is Hastie et al. (2016).

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The nonfactor, or asset-specific return on security j, $\tilde{e}_j$, is the residual return of the security after removing the estimated impacts of the K factors. The term $\tilde{f}_k$ is the return of factor "k", which is independent of securities and affects the security's return through its exposure coefficient β. The factors which can explain the security's risk best are not necessary the factors suitable for expected return modeling. They may include country, industry, and style factors. One can use two sets of risk models, fundamental or statistical, risk models to estimate Eq. (4).[8] We use the SunGard APT risk model, which uses principal components in its estimation, in this analysis.[9] The reader is referred to Blin et al. (1997) and Guerard et al. (2013) for evidence supporting the APT risk model and portfolio optimization techniques. An extensive review of factor risk models can be found in Connor et al. (2010).

Markowitz (1959) writes the general portfolio optimization objective function as:

$$minimizing\ V - \lambda * E \tag{5}$$

If the investor is more concerned about tracking a particular benchmark, the mean–variance optimization in Eq. (5) can be reformulated as a mean–variance tracking error at risk (MVTaR) optimization:

$$minimizing\left(x - x_b\right)^T C(x - x_b) - \lambda\mu^T\left(x - x_b\right) \tag{6}$$

where x_b is the weight vector of the benchmark. The total tracking error can be decomposed into the sum of systematic tracking error and the specific tracking error:

$$\begin{aligned} TE_p^2 &= (x - x_b')\beta C_{f,f}\beta'\left(x - x_b\right) + \left(x - x_b\right)'\Sigma\left(x - x_b\right) \\ &= TE_{\beta P}^2 + TE_{SP}^2 \end{aligned} \tag{7}$$

Many commercially available systems for portfolio optimization use the risk decomposition of Eq. (7). The portfolios are created with two optimization techniques; (1) traditional Markowitz mean–variance analysis as in his 1959 text, denoted MVM59; and (2) mean–variance Tracking error at Risk (MVTaR) portfolios, as risk is composed of 76% systematic risk and 24 percent specific risk, see Guerard et al. (2013). Portfolio simulations assume a 30-basis points threshold weight, a maximum stock weight of 4%, 8% monthly turnover, and a transaction cost of 150 basis points each way.

[8] The first set is a fundamental risk model, such as the Axioma World-Wide Equity Risk Factor Model (AX-WW2.1), which seeks to forecast medium-horizon risk, or risk 3–6 months ahead. The Axioma Fundamental Risk Model uses nine style factors: exchange rate sensitivity, growth (historical earnings and sales growth), leverage (debt-to-assets), liquidity (1 month trading volume divided by market capitalization), medium-term momentum (cumulative returns of the past year, excluding the previous month), short-term momentum (last month return), size (natural logarithm of issuer market capitalization), value (book-to-price and earnings-to-price ratios), and volatility (3 months average of absolute returns divided by cross-sectional standard deviation). The Axioma fundamentally-based risk model evolved from the MSCI Barra risk model. The BARRA model was developed in Rosenberg (1974), Rosenberg and Marathe (1979) and thoroughly discussed in Rudd and Clasing (1982), Grinold and Kahn (1999), Conner and Korajczyk (2010), Connor et al. (2010), and Menchero et al. (2010). Statistically-based risk models developed in the works of Ross (1976), Roll and Ross (1980), Dhrymes et al. (1984), and Guerard et al. (1997). The Axioma Statistical Risk Model, World-Wide Equity Risk Factor Model, AX-WW2.1, estimates 15 principal components to measure risk. See Guerard et al. (2015) for a comparison of Axioma Fundamental and statistically based risk models. Guerard et al. reported that the statistical model dominated the fundamental risk model in producing a higher set of returns for a given level of risk.

[9] The authors have used APT, SunGard APT, and FIS APT since 1988 and Axioma since 2010.

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5 Data mining corrections tests

We seek to maximize Geometric Means (GMs), Sharpe Ratios (ShRs), and Information Rations (IRs). We report a realized risk-return trade-off, 3/2002–5/2015 in Table 2. We report that both WLRR and LAR regression techniques produce statistically significant total active returns, risk indices return, primarily due to momentum returns, and asset selection with MVTaR technique. The WLRR and LAR regression active returns and asset selection statistics are highly statistically significant and are virtually identical with both MVM59 and MVTaR optimization techniques.[10] We use the portfolio lambda tilt of 200 which produces the highest Sharpe Ratio, see Guerard et al. (2015).

The portfolio tracking errors are substantially lower with MVTaR, as one would expect because one is minimizing the portfolio tracking error. The Information Ratio, IR, is highly for MVM59 than MVTaR in both WLRR and LAR portfolios. The asset manager and investor are compensated for bearing risk. The t-statistics on asset selection are higher in the MVM59 optimization technique. The original Markowitz mean–variance portfolio optimization technique is a very substantial benchmark.

Prudent investors are always skeptical of the great outperformance presented in marketing material, especially when the performance number is generated by back testing. Economists have long examined data mining, or data snooping, see Leamer (1978), White (1984), and Lo and Mackinlay (1990), before Markowitz and Xu (1994). The best performance number selected from multiple testing is inherently upward biased. To overcome the skepticism, we need assurance on at least two questions. First, is the "best" really the best one or it just got lucky for this test-period? Second, how much outperformance can we expected in the future from the best model if the best model is pervasively better than other models. Markowitz and Xu (1994) answered these two questions by assuming all the returns are generated by random-effect model. We now review and estimate the Markowitz and Xu (1994) data mining corrections beta.

Let y_{it} be the logarithm of one plus the return for the ith portfolio selection model in period t. The random-effect model has form

$$y_{it} = \mu_i + z_t + \varepsilon_{it} \tag{8}$$

[10] Recent research by Leamer (2016) proved highly insightful and relevant in our modeling of financial data. Leamer is critical of using t-statistic as a model selection method (dropping variables with statistically non-significant estimated coefficient) because t-value is a measure of estimation uncertainty. Leamer (2016) introduce the concept S-values to measure the sturdiness of regression coefficient taking into account the model uncertainty and use S-value as a model selection criterion. Leamer, a renowned Bayesian and data mining specialist (1973, 1978), attacks the model uncertainty by Bayesian methodology. We applied the Leamer sturdy regression technique with our data and report initial stock selection modeling enhancement. The Leamer S-regression enhances portfolio returns, the Sharpe Ratio and Information Ratios relative to the GLER model estimated with WLRR. In an initial test of WLRR and S-Regression, during the 1/2003–10/2015 time period, the WLRR Model produced a Geometric Mean of 13.75%, a Sharpe Ratio of 0.738, an IR of 1.08, and a Specific Return of 4.23% (t-statistic of 3.70). The corresponding portfolio statistics for the Leamer Sturdy-regressions were a Geometric Mean of 15.85%, a Sharpe Ratio of 0.748, an IR of 1.16, and a Specific Return of 6.27% (t-statistic of 4.54). The WLRR Model produced a tracking error of 7.30% whereas the Leamer Sturdy-regression Model produced a 9.53% tracking error. Clearly, on the basis of the Geometric Mean, Sharpe ratio, and the Information Ratio, the Leamer Sturdy-regression model is well worth further research.

The authors Markowitz and Guerard (2019), presented a paper, "The Existence and Persistence of Financial Anomalies" at the Q-Group, The Institute of Quantitative Research meeting, October 29, 2019 in La Jolla. The authors report additional research on the optimal influence function regression modeling in Maronna et al. (2019) enhances the Geometric Mean, Sharpe Ratio, and Information Ratio relative to WLRR.

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Table 2 Mean–variance optimized portfolios

Universe: global, two-analyst, top 7500 stock portfolios, December 2002–May 2015							
Optimizer: APT							
Lambda is 200							
Attribution: Barra							
Optimization		Active	Act. eq.	Risk	Risk ind.	Asset	Asset sel.
Technique	Model	Equity (%)	T-stat	Indices (%)	T-stat	Selection (%)	T-stat
MVM59	WLRR	20.14	6.01	10.89	5.82	5.59	5.59
	LAR	19.27	5.98	11.24	6.05	5.33	5.21
MVTaR	WLRR	14.44	5.58	9.53	6.40	4.76	3.04
	LAR	14.46	5.35	9.45	6.30	4.87	2.85
Optimization		Tracking	Trans.	Total	T. active	Information	
Technique	Model	Error, TE	Costs (%)	Active (%)	T-stat	Ratio	
MVM59	WLRR	11.46	5.06	17.75	5.28	1.45	
	LAR	11.39	4.99	16.99	5.16	1.42	
MVTaR	WLRR	9.06	4.09	11.35	4.43	1.33	
	LAR	9.48	5.26	10.49	3.95	1.08	

where μ_i is the model effect, z_t is the period effect, and ϵ_{it} is the random deviation. These effects are all assumed to be uncorrelated. The appropriate estimate of μ_i is not its sample average return

$$\bar{r}_i = \frac{\sum_{t=1}^{T} y_{it}}{T} \tag{9}$$

but rather

$$\hat{\mu}_i = \bar{r} + \beta\left(\bar{r}_i - \bar{r}\right) \tag{10}$$

where $\bar{r} = \sum_{i=1}^{N} \bar{r}_i / N$ is the grand mean of methods and $\beta = \frac{\mathrm{Cov}(\bar{r}_i, \mu)}{\mathrm{Var}(\bar{r}_i)}$ is the regression coefficient of μ_i as a function of $\bar{r}_i$ such that $0 < \beta < 1$. In other word, the best estimate of model effect μ_i is its sample estimate regressed back to the average estimate (the grand average). We report that the global universe data mining beta of Table 1 models is 0.68, with a F-statistic of 2.98, and is highly statistically significant.[11] The Markowitz and Xu (1994) test results imply that one should expect 68% of the model excess returns to be continued. There is no evidence of data mining in global stocks. Guerard et al. (2019) report the original Markowitz and Xu (1994) data mining corrections (DMC) test as a "Regression to the Mean" test. We now introduce the reader to more modern data mining tests.

[11] Had the authors restricted the DMC test to the GPRD variables, the DMC beta would have been 0.70 and its corresponding t-statistic equal to 3.01. Thus, we report little difference in the DMC results of 21 and 36 models tested in the global universe.

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5.1 Additional evidence on single performance measurement and testing

Performance measurements like the Treynor Index, Jensen's alpha, and the Sharpe Ratio were introduced in Guerard et al. (2019, Chapter 5), and more modern approaches to data mining corrections were introduced in Chapter 6. These measurements are point estimates. We will study the estimation error associated with these measurements to get a range of estimates which enable us to test against the null hypothesis. The statistic of Sharpe Ratio is from Lo (2002). The statistic of Treynor index is new material and is derived with generalized methods of moments (GMM) estimation technique of Hansen (1982).

Assume that we have found an attractive investment model generating portfolio P. The portfolio's geometric mean can be approximated in terms of mean and variance, see Markowitz (1959, Chapter 5):

$$gm \approx \mu - 0.5 \cdot \sigma^2 \tag{11}$$

where (μ, σ) are the portfolio's expected return and standard deviation. Let $R_{p,t}$ be the portfolio's return at period t less or equal to total period T. The estimation of (μ, σ) from sample returns is

$$\hat{\mu} = \frac{1}{T}\sum_{t=1}^{T} R_{p,t}$$

$$\hat{\sigma}^2 = \frac{1}{T-1}\sum_{t=1}^{T} (R_{p,t} - \hat{\mu})^2.$$

The estimated geometric mean is

$$\widehat{gm} = \hat{\mu} - 0.5 \cdot \hat{\sigma}^2.$$

It is easy to check that $\hat{\mu}$ and $\hat{\sigma}^2$ are not correlated. By the property of Chi squared distribution the variance of $\hat{\mu}$ and $\hat{\sigma}^2$ are:

$$\text{var}(\hat{\mu}) = \sigma^2/T$$

and

$$\text{var}(\hat{\sigma}^2) = 2\sigma^4/(T-1).$$

These equations imply that

$$\sigma_{\widehat{gm}} \sim \sqrt{\text{var}(\hat{\mu}) + 0.5 \cdot 0.5\text{var}(\hat{\sigma}^2)} \approx \frac{\sigma\sqrt{1+\sigma^2}}{\sqrt{T}} \approx \frac{\hat{\sigma}\sqrt{1+\hat{\sigma}^2}}{\sqrt{T}} \tag{12}$$

Geometric mean could be considered be a risk-adjusted measure of performance. The Sharpe ratio is the most popular risk-adjusted performance measure defined as:

$$SR = \frac{\mu - R_f}{\sigma} \tag{13}$$

where R_f is the risk-free rate. Most portfolios' annual Sharpe ratios range from 0 to 1. Higher sharpe Ratios are preferred. The estimated Sharpe Ratio is

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$$\widehat{SR} = \frac{\hat{\mu} - R_f}{\hat{\sigma}}$$

Lo (2002) use GMM technique to derive the estimation error

$$\sqrt{T}(\widehat{SR} - SR) \approx N(0, V_{IID}),$$

where

$$V_{IID} = 1 + \frac{(\mu - R_f)^2}{2\sigma^2} = 1 + 0.5 * SR^2 \tag{14}$$

Lo also derives the formula for the Sharpe ratio and its estimation error in the presence of serial correlations in the returns.

Other performance measurements are measured against benchmarks. It is natural to compare the portfolio's return versus benchmark's return. Let $R_{b,t}$ be the benchmark return at period t less than total period T. Assume that the return difference $R_{p,t} - R_{b,t}$ be i.i.d. with distribution $N(\mu, \sigma^2)$.

The sample mean and standard deviation of return differences are

$$\hat{\mu} = \frac{1}{T}\sum_{t=1}^{T}(R_{p,t} - R_{b,t})$$

$$\hat{\sigma} = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(R_{p,t} - R_{b,t} - \hat{\mu})^2}$$

The standard t-statistic is

$$t = \frac{\sqrt{T} * \hat{\mu}}{\hat{\sigma}} \tag{15}$$

which can be used to test the null hypothesis $\mu = 0$ (Table 3).

Jensen alpha is the regression intercept of portfolio excess returns against market excess returns.

$$R_{p,t} - r_{f,t} = \alpha_p + \beta_p(R_{m,t} - r_{f,t}) + \varepsilon_t$$

The t-statistic from the standard regression analysis is

$$t = \frac{\widetilde{\alpha}}{std.err(\widetilde{\alpha})} \tag{16}$$

which can be used to test the null hypothesis

$$\alpha_p = 0$$

The earliest market risk adjusted performance measurement is the Treynor index.

$$TI = \frac{(\mu - r_f)}{\beta} = \frac{(\mu - r_f) * v_m}{c_{pm}}$$

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Table 3 Monthly excess returns for March 2002–May 2015

	Average	SD	t-value	p value
LAR	0.79	2.5	3.97	0.00011
WLAR	0.80	2.55	3.96	0.00011
WLRR	0.75	2.6	3.65	0.00035
CTEF	0.65	2.41	3.41	0.00084
LASSO	0.67	2.58	3.29	0.00125
SP	0.69	3.07	2.85	0.00495
EP	0.54	2.96	2.29	0.02327
CTEFROIC	0.36	2.66	1.7	0.09038
BR1	0.12	0.93	1.65	0.10113
ROE1YR	0.18	1.57	1.46	0.14724
FEP1	0.41	3.64	1.41	0.16047
FEP2	0.39	3.66	1.34	0.18241
ROIC	0.17	1.82	1.19	0.23677
PMTREND	0.21	2.47	1.06	0.29017
ALPHA	0.28	3.39	1.06	0.29066
CP	0.26	3.14	1.05	0.29724
PM71	0.25	3.06	1.04	0.30172
BR2	0.08	0.95	1.03	0.30474
CSR	0.14	1.96	0.92	0.3565
ES	0.17	2.34	0.91	0.3649
NCSR	0.12	1.88	0.84	0.40417
ROA1YR	0.1	1.82	0.73	0.46907
NDR	0.12	2.4	0.66	0.51122
ROE3YR	0.07	1.65	0.51	0.60786

where v_m is the variance of the market return and c_{pm} is the covariance of the portfolio return with the market return. Denote $\theta \equiv (\mu - r_f, \mu_m - r_f, v_p, v_m, c_{pm})'$ the vector of parameters to be estimated, where the additional parameter μ_m is the market's expected returns and v_p is the portfolio variance σ^2. The moments used in GMM are the means and variances and covariance for the portfolio and markets. They are defined as

$$\begin{aligned}
\varphi_1(x_t, \theta) &= R_{p,t} - \mu \\
\varphi_2(x_t, \theta) &= R_{m,t} - \mu_m \\
\varphi_2(x_t, \theta) &= (R_{p,t} - \mu)^2 - v_p \\
\varphi_4(x_t, \theta) &= (R_{m,t} - \mu_m)^2 - v_m \\
\varphi_5(x_t, \theta) &= (R_{p,t} - \mu)(R_{m,t} - \mu_m) - c_{pm}
\end{aligned}$$

Denote moments function vector $\varphi = (\varphi_1, \ldots, \varphi_5)'$ and return vector $x_t = (R_{p,t}, R_{m,t})'$. The GMM estimator of θ, denoted by $\hat{\theta}$, is given by the solution to moments conditions

$$\frac{1}{T} \sum_{t=1}^{T} \varphi(x_t, \hat{\theta}) = 0 \tag{17}$$

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Hansen (1982) shows that

$$\sqrt{T}(\hat{\theta}-\theta)\sim N(0,V_\theta), V_\theta \equiv H^{-1}\Sigma H^{-1} \tag{18a}$$

where

$$H \equiv \lim_{T\to\infty} E\left[\frac{1}{T}\sum_{t=1}^{T}\varphi_\theta(X_t,\theta)\right] \tag{18b}$$

and

$$\Sigma \equiv \lim_{T\to\infty} E[\frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{T}\varphi(X_t,\theta)\varphi'(X_s,\theta)] \tag{18c}$$

It turns out that H is a very simple matrix and Σ is easy to calculate. By definition of φ,

$$\varphi_\theta(X_t,\theta)=\begin{pmatrix} -1 & 0 & 0 & 0 & 0\\ 0 & -1 & 0 & 0 & 0\\ -2(R_{p,t}-\mu) & 0 & -1 & 0 & 0\\ 0 & -2(R_{m,t}-\mu_m) & 0 & -1 & 0\\ R_{m,t}-\mu_m & R_{p,t}-\mu & 0 & 0 & -1 \end{pmatrix}$$

This implies that

$$H=-I \tag{19}$$

The reader can check that

$$\Sigma=\begin{pmatrix} v_p & c_{pm} & & 0 & \\ c_{pm} & v_m & & & \\ & & 2v_p^2 & 2c_{pm}^2 & 2v_pc_{pm}\\ 0 & & 2c_{pm}^2 & 2v_m^2 & 2v_mc_{pm}\\ & & 2v_pc_{pm} & 2v_mc_{pm} & v_pv_m+c_{pm}^2 \end{pmatrix} \tag{20}$$

Any function of parameter vector θ, $g(\theta)$, can be estimated by the delta method:

$$\sqrt{T}(g(\hat{\theta})-g(\theta))\sim N(0,V_g), V_g \equiv \frac{\delta g}{\delta\theta}\Sigma\frac{\delta g}{\delta\theta'} \tag{21}$$

The H matrix is eliminated in Eq. (21) because of Eq. (19). By applying the delta method to the Treynor index, we have

$$\frac{\delta g}{\delta\theta}=\begin{pmatrix} \frac{v_m}{c_{pm}} & 0 & 0 & \frac{\mu_p-r_f}{c_{pm}} & -\frac{(\mu_p-r_f)v_m}{c_{pm}^2} \end{pmatrix}$$

A tedious calculation shows

$$V_g=\frac{v_pv_m^2}{c_{pm}^2}+\frac{(\mu_p-r_f)^2v_m^2}{c_{pm}^2}\left(\frac{v_pv_m}{c_{pm}^2}-1\right)=\frac{v_m}{\rho^2}+(TI)^2\left(\frac{1}{\rho^2}-1\right) \tag{22}$$

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where $\rho = \frac{c_{pm}}{\sqrt{v_p v_m}}$ is the correlation coefficient between the portfolio returns and the market returns.

5.2 Multiple hypothesis testing and false discovery rate

Harvey and Liu (2014), Harvey et al. (2016), and Harvey (2017) discuss and test more recent data mining corrections test in the tradition of the Family-Wise Error Rate (FWER) test, in which it is unacceptable to make a (single) false discovery. In a single, independent, a two-sigma test, the chance of a single false discovery is only 5%. The best known FWER test is the Bonferroni test. A second FWER test is the Holm method begins by sorting the tests from the lowest p value (most significant) to the highest (least significant) and captures the information in the distribution of the test statistics. The Holm test is less stringent than the Bonferroni because the hurdles are relaxed after the first test. The False Discovery Rate approach allows an expected proportional error rate, see Benjamini and Hochberg (1995) and Benjamini and Yekutieli (2001).[12] Harvey and Liu (2014) denoted Benjamini and Hockberg test as the BHY procedure. We apply the BHY test and report very similar results to GLER data in Guerard et al. (2014), finding that the WLRR, LAR variables were not "lucky", or data mined factors, see Table 4 for MVM59 results and Table 5 for MVTaR results.

The BHY test results are very consistent when tested with MVM59 and MVTaR optimization techniques.

The Markowitz and Xu (1994) data mining corrections test is consistent with the Zhou et al. (2006) FDR application and the Harvey et al. (2016) FWER test.

We described the most used performance measurements and their error estimations in portfolio management. The error estimation leads us to calculate t-statistics, infer the p value, which gives us the statistical confidence of assessing the performance of each model. We list the statistic t_i in Tables 4 and 5 and the corresponding p value p_i for each investment model using excess return the as performance measurement. The robust least angular regression (WLAR) has the highest t-statistic among the models listed with $t_{\max} = \max\{t_1, t_2, \ldots, t_n\} = 3.97$ and p value 0.00011. We can reject the null hypothesis that LAR has no excess return. The error of rejecting this hypothesis is less than 0.01 percent. The maximum t-statistic t_{RLAR} is inflated by chance, obviously. If t_i are IIDs, then

$$P(t_{\max} > t) = 1 - P(t_{\max} \leq t) = 1 - \prod_{i=1}^{n} P(t_i \leq t) = 1 - \prod_{i=1}^{n} [1 - P(t_i > t)]$$

If we choose t_i such that $P(t_i > t) = \alpha$. i.e., we control the type I error to be less or equal than α for each model. The type I error for the luckiest model is

$$P(t_{\max} > t) = 1 - \prod_{i=1}^{n} [1 - P(t_i > t)] = 1 - (1 - \alpha)^n = \overline{\alpha}$$

[12] Zhou et al. (2006) discussed the role of alpha-investing as a form of Information Investing to minimize the FDR. The alpha-investing approach uses p values associated with t-statistics to represent the probability that coefficients could have been estimated by chance even though the true coefficients were zero.

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Table 4 BHY false discovery rate test, March 2002–May 2015

Optimization technique is mean–variance (MVM59)								
Portfolio Lambda is 200								
Name	Mean	SD	IR	34	t	C(M)	adjust IR	HairCut
WLRR	1.4644	3.3202	0.4411	1.0000	5.5615	0.0000	0.3486	0.2096
LASSO	1.3103	3.1313	0.4185	0.5000	5.2765	0.0000	0.3481	0.1681
RWLAR	1.3964	3.3455	0.4174	0.3333	5.2630	0.0000	0.3481	0.1659
LAR	1.3457	3.3423	0.4026	0.2500	5.0769	0.0000	0.3389	0.1582
CTEF	1.1207	2.9019	0.3862	0.2000	4.8699	0.0000	0.3265	0.1545
EP	0.9133	3.5436	0.2577	0.1667	3.2498	0.0009	0.1863	0.2770
FEP1	1.0943	4.5039	0.2430	0.1429	3.0637	0.0016	0.1734	0.2865
SP	0.8425	3.5423	0.2378	0.1250	2.9991	0.0020	0.1731	0.2721
FEP2	1.1824	5.0145	0.2358	0.1111	2.9733	0.0021	0.1731	0.2658
CTEFROIC	0.7110	3.1042	0.2290	0.1000	2.8882	0.0027	0.1686	0.2641
PMTREND	0.6278	3.3867	0.1854	0.0909	2.3374	0.0114	0.1171	0.3681
ROE1YR	0.2799	1.6146	0.1734	0.0833	2.1860	0.0164	0.1084	0.3747
CP	0.6737	3.8885	0.1733	0.0769	2.1847	0.0164	0.1084	0.3744
ROIC	0.2886	1.7527	0.1646	0.0714	2.0761	0.0211	0.1003	0.3909
BP	0.6984	4.7539	0.1469	0.0667	1.8524	0.0344	0.0793	0.4604
PM71	0.5324	3.9543	0.1346	0.0625	1.6978	0.0474	0.0652	0.5160
NCSR	0.3580	2.9177	0.1227	0.0588	1.5474	0.0635	0.0509	0.5849
CSR	0.3504	2.9605	0.1184	0.0556	1.4925	0.0704	0.0508	0.5710
ROE3YR	0.1943	1.6480	0.1179	0.0526	1.4867	0.0712	0.0508	0.5693
FY1RV3	0.3054	2.7570	0.1108	0.0500	1.3967	0.0838	0.0433	0.6088
ROE5YR	0.1667	1.5943	0.1046	0.0476	1.3184	0.0962	0.0376	0.6401
ALPHA	0.4018	4.1353	0.0972	0.0455	1.2251	0.1127	0.0376	0.6127
ROA1YR	0.1721	1.8516	0.0930	0.0435	1.1722	0.1229	0.0376	0.5952
ROA3YR	0.1203	1.8422	0.0653	0.0417	0.8235	0.2068	0.0376	0.4238
STDEV	0.1225	3.1019	0.0395	0.0400	0.4982	0.3101	0.0376	0.0474
ROA5YR	0.0641	1.8162	0.0353	0.0385	0.4449	0.3290	0.0376	− 0.0665
ES	0.1015	3.2071	0.0317	0.0370	0.3992	0.3456	0.0376	− 0.1888
NDR	0.0783	3.1702	0.0247	0.0357	0.3114	0.3783	0.0376	− 0.5238
PM1	0.0922	3.8876	0.0237	0.0345	0.2992	0.3829	0.0376	− 0.5861
DR	0.0678	3.8742	0.0175	0.0333	0.2205	0.4131	0.0376	− 1.1519
DI	0.0704	5.4405	0.0129	0.0323	0.1631	0.4355	0.0376	− 1.9093
BR1	0.0155	1.6062	0.0097	0.0313	0.1219	0.4517	0.0376	− 2.8914
BR2	− 0.0208	1.5545	− 0.0134	0.0303	− 0.1686	0.5667	0.0376	3.8140
CSI	− 0.1114	3.9957	− 0.0279	0.0294	− 0.3517	0.6369	0.0376	2.3492

In case $\alpha = 0.05$ with the corresponding t = 2 and n = 36, $P(t_{\max} > t) = 0.84$ i.e., the probability of a type I error is almost 85%. To have $\overline{\alpha}$ less than 5%, we need to choose α to be 0.000145 and the corresponding t-statistic must be at least 3.19. We reject the null hypothesis of data mining in both the Markowitz and Xu (1994) data mining corrections test and the BHY false discovery test.

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Table 5 BHY false discovery rate test, march 2002–May 2015

Optimization technique is mean–variance tracking error at risk (MVTaR)								
Portfolio Lambda is 200								
Name	Mean	SD	IR	t	C(M)	4.0580	Adjust IR	HairCut
LAR	0.9037	2.6985	0.3349	4.2226	0.0000	0.0054	0.2290	0.3163
WLRR	0.8625	2.8041	0.3076	3.8784	0.0001	0.0064	0.2239	0.2721
LAR	0.8567	2.8123	0.3046	3.8411	0.0001	0.0064	0.2239	0.2651
LASSO	0.7302	2.8170	0.2592	3.2685	0.0009	0.0291	0.1773	0.3159
CTEF	0.6418	2.5553	0.2512	3.1673	0.0012	0.0314	0.1747	0.3044
SP	0.7121	3.1842	0.2236	2.8201	0.0032	0.0703	0.1461	0.3466
EP	0.6238	3.2238	0.1935	2.4401	0.0088	0.1638	0.1118	0.4222
CTEFROIC	0.5476	2.8974	0.1890	2.3829	0.0102	0.1652	0.1114	0.4105
FEP1	0.5368	3.8853	0.1382	1.7422	0.0433	0.6248	0.0390	0.7178
BR2	0.1198	0.9450	0.1268	1.5988	0.0576	0.6964	0.0311	0.7548
FEP2	0.4913	3.9955	0.1230	1.5506	0.0631	0.6964	0.0311	0.7471
PMTREND	0.3211	2.6564	0.1209	1.5241	0.0664	0.6964	0.0311	0.7427
BR1	0.1221	1.0415	0.1172	1.4780	0.0723	0.6964	0.0311	0.7347
ROIC	0.1913	1.8129	0.1055	1.3308	0.0941	0.6964	0.0311	0.7054
ROE1YR	0.1682	1.6309	0.1031	1.3004	0.0992	0.6964	0.0311	0.6985
BP	0.5120	5.0592	0.1012	1.2762	0.1034	0.6964	0.0311	0.6928
ALPHA	0.3110	3.7258	0.0835	1.0526	0.1484	0.6964	0.0311	0.6275
PM71	0.2555	3.4445	0.0742	0.9352	0.1767	0.6964	0.0311	0.5808
CP	0.2526	3.4429	0.0734	0.9253	0.1793	0.6964	0.0311	0.5763
ROE3YR	0.1154	1.6423	0.0702	0.8857	0.1897	0.6964	0.0311	0.5573
ES	0.1539	2.2667	0.0679	0.8562	0.1977	0.6964	0.0311	0.5421
CSR	0.1364	2.1977	0.0621	0.7827	0.2184	0.6964	0.0311	0.4991
ROA1YR	0.1130	1.8835	0.0600	0.7562	0.2262	0.6964	0.0311	0.4815
NCSR	0.0618	2.0589	0.0300	0.3787	0.3531	0.6964	0.0311	− 0.0354
DI	0.1136	4.5053	0.0252	0.3179	0.3758	0.6964	0.0311	− 0.2333
STDEV	0.0381	2.2240	0.0171	0.2161	0.4148	0.6964	0.0311	− 0.8141
FY2RV3	0.0440	3.3410	0.0132	0.1659	0.4344	0.6964	0.0311	− 1.3627
ROA3YR	− 0.0062	1.8670	− 0.0033	− 0.0420	0.5167	0.6964	0.0311	10.3346
ROE5YR	− 0.0170	1.5796	− 0.0107	− 0.1355	0.5537	0.6964	0.0311	3.8936
PM1	− 0.0685	2.9299	− 0.0234	− 0.2947	0.6154	0.6964	0.0311	2.3303
FY1RV3	− 0.0895	2.5062	− 0.0357	− 0.4502	0.6729	0.6964	0.0311	1.8708
ROA5YR	− 0.0754	1.8399	− 0.0410	− 0.5168	0.6964	0.6964	0.0311	1.7587

6 Summary and conclusions

In this analysis, we report evidence confirming the continued relevance of the robust regression models in estimating global stock selection models. The Beaton–Tukey weighted latent root regression technique continues to produce statistically significant results in the 2003–2015 time period, passing information coefficient and Data Mining Corrections tests in global universe. We believe WLRR model is close to the ideal expected return

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model because the model has continued to produce statistically significant active returns and active returns and positive specific returns, post-publication, 1993. The WLRR model can be used with sophisticated risk models and constrained mean–variance optimization processes to exploit market inefficiency. We report that (1) robust regression applications are appropriate for modeling stock returns in global markets; (2) weighted latent root regression robust regression techniques work as well as LAR and LASSO-Regressions in building effective stock selection models; (3) mean–variance techniques continue to produce portfolios capable of generating excess returns above transactions costs; and (4) our models pass several data mining tests such that regression models produce statistically significant asset selection for global stocks. The primary contributions of this analysis are to test several modern regression techniques and modern (2015–2018) data mining and false discovery rate modeling techniques. Recent Sturdy-Regression modeling technique may offer the greatest potential for further research for statistically based stock selection modeling.

Compliance with ethical standards

Conflict of interest The views and opinions expressed in this paper are those of the authors and may not represent or reflect those of McKinley Capital Management, LLC. All information contained herein is believed to be acquired from reliable sources but the accuracy cannot be guaranteed. This paper is for informational purposes only; was prepared for academics and financially sophisticated and institutional audience; and does not represent specific financial services or investment recommendations or advice.

References

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B, 57,* 289–300.

Benjamini, Y., & Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics, 2,* 1165–1188.

Blin, J. M., Bender, S., & Guerard, J. B., Jr. (1997). Earnings forecasts, revisions and momentum in the estimation of efficient market-neutral Japanese and U.S. portfolios. *Research in Finance, 15,* 93–114.

Bloch, M., Guerard, J. B., Jr., Markowitz, H. M., Todd, P., & Xu, G. (1993). A comparison of some aspects of the U.S. and Japanese equity markets. *Japan & the World Economy, 5,* 3–26.

Conner, G., & Korajczyk, R. A. (2010). Factor models in portfolio and asset pricing theory. In J. Guerard (Ed.), *The handbook of portfolio construction: Contemporary applications of Markowitz techniques*. New York: Springer.

Connor, G., Goldberg, L., & Korajczyk, R. A. (2010). *Portfolio risk analysis*. Princeton: Princeton University Press.

Deng, S., & Min, X. (2013). Applied optimization in global efficient portfolio construction using earnings forecasting. *Journal of Investing, 22,* 104–114.

Dhrymes, P. J. (2017). *Introductory econometrics* (2nd ed.). New York: Springer.

Dhrymes, P. J., Friend, I., & Gultekin, N. B. (1984). A critical re-examination of the empirical evidence on the arbitrage pricing theory. *Journal of Finance, 39,* 323–346.

Efron, B., Hastie, T., Johnstone, J., & Tibshirani, R. (2004). Least angle regression. *The Annals of Statistics, 32,* 407–499.

Elton, E. J., Gruber, M. J., Brown, S. J., & Goetzman, W. N. (2007). *Modern portfolio theory and investment analysis* (7th ed.). New York: Wiley.

Fama, E. F., & French, K. R. (1992). Cross-sectional variation in expected stock returns. *Journal of Finance, 1992*(47), 427–465.

Fama, E. F., & French, K. R. (1995). Size and the book-to-market factors in earnings and returns. *Journal of Finance, 50,* 131–155.

Author's personal copy

Fama, E. F., & French, K. R. (2008). Dissecting anomalies. *Journal of Finance, 63,* 1653–1678.
Farrar, D. E., & Glauber, R. (1967). Multicollinearity in regression analysis; The problem revisited. *The Review of Economics and Statistics, 49,* 92–107.
Graham, B., & Dodd, D. (1934). *Security analysis: Principles and technique*. New York: McGraw-Hill Book Company.
Grinold, R., & Kahn, R. (1999). *Active portfolio management*. New York: McGraw-Hill/Irwin.
Guerard, J. B., Jr., Gillam, R. A., Markowitz, H. M., Xu, G., Deng, S., & Wang, Z. (2018a). Data mining corrections testing in Chinese stocks. *Interfaces, 48,* 108–120.
Guerard, J. B., Jr., Gultekin, M., & Stone, B. K. (1997). The role of fundamental data and analysts' earnings breadth, forecasts, and revisions in the creation of efficient portfolios. *Research in Finance, 15,* 69–92.
Guerard, J. B., Jr., Markowitz, H. M., & Xu, G. (2014). The role of effective corporate decisions in the creation of efficient portfolios. *IBM Journal of Research and Development, 58*(6), 1–11.
Guerard, J. B., Jr., Markowitz, H. M., & Xu, G. (2015). Earnings forecasting in a global stock selection model and efficient portfolio construction and management. *International Journal of Forecasting, 31,* 550–560.
Guerard, J. B., Jr., Markowitz, H. M., Xu, G., & Wang, E. (2018b). Global portfolio construction with emphasis on conflicting corporate strategies to maximize stockholder wealth. *Annals of Operations Research, 267,* 203–219.
Guerard, J. B., Jr., Rachev, R. T., & Shao, B. (2013). Efficient global portfolios: Big data and investment universes. *IBM Journal of Research and Development, 57,* 11.
Guerard, J. B., Jr., Xu, G., & Gultekin, M. N. (2012). Investing with momentum: The past, present, and future. *Journal of Investing, 21,* 68–80.
Guerard, J. B., Jr., Xu, G., & Wang, Z. (2019). *Portfolio and investment analysis with SAS*. Cary, NC: SAS Press.
Gunst, R. F., & Mason, R. L. (1980). *Regression analysis and its application*. New York: Marcel Dekker Inc.
Hansen, L. (1982). Large sample properties of generalized method of moments estimator. *Econnometrica, 50*(4), 1029–1054.
Harvey, C. R. (2017). Presidential address: The scientific outlook in financial economics. *Journal of Finance, 72,* 1399–1440.
Harvey, C. R. & Liu, Y. (2014). Lucky Factors. SSRN. http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2528780.
Harvey, C. R., Liu, Y., & Zhu, H. (2016). …and the cross-section of expected returns. *Review of Financial Studies, 29*(1), 5–69.
Hastie, T., Tibshirani, R., & Friedman, J. (2016). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed., 11th printing). New York: Springer.
Haugen, R. A. (2001). *Modern investment theory* (5th ed.). Upper Saddle River, NJ: Prentice Hall.
Haugen, R. A., & Baker, N. (1996). Communality in the determinants of expected results. *Journal of Financial Economics, 41,* 401–440.
Haugen, R. A., & Baker, N. (2010). Case closed. In J. B. Guerard (Ed.), *The Handbook of portfolio construction: Contemporary applications of Markowitz techniques*. New York: Springer.
Jacobs, B., & Levy, K. (1988). Disentangling equity return regularities: New insights and investment opportunities. *Financial Analysts Journal, 44,* 18–43.
Leamer, E. E. (1973). Multicollinearity: A Bayesian interpretation. *Review of Economics and Statistics, 1973*(55), 371–380.
Leamer, E. E. (1978). *Specification searches: Ad hoc inference with nonexperimental data* (p. 1978). New York: Wiley.
Leamer, E. E. (2016). S-values; conventional context-minimal measures of the sturdiness of regression coefficients. *Journal of Econometrics, 19,* 147–161.
Levy, H. (1999). *Introduction to investments* (2nd ed.). Cincinnati: South-Western College Publishing.
Levy, H. (2012). *The capital asset pricing model in the 21st century*. New York: Cambridge University Press.
Lin, D., Foster, D. P., & Ungar, L. H. (2011). VIF regression: A fast regression algorithm for large data. *Journal of the American Statistical Association, 10,* 232–247.
Lo, A. (2002). The statistics of Sharpe ratios. *Financial Analyst Journal, 58*, 36–52.
Lo, A., & MacKinlay, C. (1990). Data-snooping biases in tests of financial asset pricing models. *The Review of Financial Studies, 3,* 431–467.
Markowitz, H. M. (1952). Portfolio selection. *Journal of Finance, 7,* 77–91.

Author's personal copy

Markowitz, H. M. (1959). *Portfolio selection: Efficient diversification of investment.* Cowles Foundation Monograph No. 16. New York: Wiley.
Markowitz, H. M., & Guerard, J. B. Jr. (2019). *The existence and persistence of financial anomalies*. To be presented at The Institute of Quantitative research in Finance. La Jolla, October.
Markowitz, H. M., & Xu, G. (1994). Data mining corrections. *Journal of Portfolio Management, 21,* 60–69.
Maronna, R. A., Martin, R. D., & Yohai, V. J. (2006). *Robust statistics: Theory and methods*. New York: Wiley.
Maronna, R. A., Martin, R. D., Yohai, V. J., & Salibian-Barerra, M. (2019). *Robust statistics: Theory and methods*. New York: Wiley.
Menchero, J., Morozov, A., & Shepard, P. (2010). Global Equity Modeling. In J. B. Guerard (Ed.), *The handbook of portfolio construction: Contemporary applications of Markowitz techniques*. New York: Springer.
Ramnath, S., Rock, S., & Shane, P. (2008). The financial analyst forecasting literature: A taxonomy with suggestions for further research. *International Journal of Forecasting, 24,* 34–75.
Roll, R., & Ross, A. (1980). An empirical investigation of the arbitrage pricing theory. *Journal of Finance, 35,* 1071–1103.
Rosenberg, B. (1974). Extra-market components of covariance in security returns. *Journal of Financial and Quantitative Analysis, 9,* 263–274.
Rosenberg, B., & Marathe, V. (1979). Tests of capital asset pricing hypotheses. In H. Levy (Ed.), *Research in finance* (Vol. 1). Greenwich, CT: JAI Press.
Ross, S. A. (1976). The arbitrage theory of capital asset pricing. *Journal of Economic Theory, 13,* 341–360.
Rudd, A., & Clasing, H. K. (1982). *Modern portfolio theory: The principles of investment management*. Homewood, IL: Dow-Jones Irwin.
Subramanian, S., Suzuki, S. D., Makedon, A., Hall, J., Pouey, M., & Wang, B. (2018). *A pm's guide to stock picking*. Bank of America Merrill Lynch.
Tibshirani, R. (1996). Regression shrinkage and selection via the LASSO. *Journal of the Royal Statistical Association, Series B, 1996*(58), 267–288.
White, H. (1984). *Asymptotic theory for econometricians*. New York: Academic Press.
Williams, J. B. (1938). *The theory of investment value*. Cambridge, MA: Harvard University Press.
Zhou, J., Foster, D. P., Stine, R. A., & Ungar, L. H. (2006). Streamwise feature selection. *Journal of Machine Learning Research, 2006*(7), 1861–1885.
Ziemba, W. T., & Schwartz, S. (1993). *Invest Japan*. Chicago: Probus.

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